

**Wearable Bike**

Behold the wearable bike concept by Yamaha. The bike allows two riding positions

**Mission 1**

An electric bike that is capable at reaching 150 miles per hour, runs on lithium-ion batteries

**Yamaha EC2**

This may look like a bike, but it is actually a two-wheeled iPod dock. Yes, you can ride it to work

pretty much made it a standard feature on all their top-of-the-line road-going machines. And why not? It was the first to master the art of using traction control effectively in MotoGP, so what you buy today is based on the exact technology that helped Casey Stoner win his world championship in 2007. In theory, the traction control system is rather simple and seems like it was sketched up on the back of a napkin in a coffee shop – if one wheel is rotating faster than the other, use some means to slow it down and recover lost traction. In practice though, the system relies on on-board gyros, accelerometers, gear box position sensors, and RPM sensors, to feed inputs to the ECU which are processed instantaneously as the system tries to slow one wheel down. The slowing down process can happen in a number of ways and usually differs with each manufacturer and the sophistication of the system. In the early 90's Honda had a primitive version of traction control where if one wheel was spinning faster than the other, the ECU applied the brakes on the faster wheel in order to reduce speed. Since then, the system has improved in leaps and bounds. Yamaha's MotoGP bike, the YZF-M1, uses an assortment of sensors and a GPS system to detect the spin and controls it by partially shutting off the throttle. Ducati, on the other hand, goes about it in a different way. The ECU on the Ducati knows the grip the tyre and the track have on offer at various temperatures



and lean angles and prevents the spin altogether by limiting the amount of power that goes to the rear wheel. Many more variables to work with and a little more complex, but so much more effective than the competition that a little Australian kid has been running circles around an eight-times world champion.

Stop and go
Anti-lock braking system

Electronic advancements have not been restricted to engines and engine management, but have also found their way into what seemed to be uncomplicated parts on the chassis as well. Brakes and suspension have also been hooked up to electronics. Honda's advanced braking technology, the Combined ABS, has made its way to



India this year on board the CBR1000RR. So far, ABS, which is synonymous with cars, has been seldom used on motorcycles. Developmental systems were used on motorcycles, none of which were as fast or as sophisticated, or demanded a high threshold for braking, or brake performance, as the current day crop. However, along with other advances, ABS has also come of age. Honda's been busy developing what is known as Combined ABS for a few years now, to make its motorcycles safer to ride. The system, unlike conventional ABS, uses

brake-by-wire technology that senses the force applied to the brake lever. This is relayed to the central processor which in-turn instructs motors on how much hydraulic force to send to both the front and the rear brake calipers, depending on your speed and the degree of deceleration. This means that a novice rider on a motorcycle equipped with Combined ABS can now grab a handful of lever and forgivingly come to a halt without having to think about, or risk, a low-side, and writing off a couple of lakhs in a crash. Racers and journalists the world over claim that the

SIDEBURN

Traction control was still in its nascent stage in the 90s and that was when motorcycle racing was at its lethal best. Rear wheels trying to overtake the front were the norm and a high-side wasn't a term used to describe a rider's mood after he won a race. Racers had nothing but pure skill and handlebars to hold on to. <http://tinyurl.com/pxf5xb>



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